Assignment 11 — Process Management

Subject: Operating Systems
Assignment No.: 11
Title: Process Management and Scheduling
Difficulty: Intermediate
Submission Format: PDF / DOCX

Objective

To understand processes, process states, scheduling, and common CPU scheduling algorithms.

Questions

1. What is a process?
2. Explain the difference between a program and a process.
3. Explain the process life cycle.
4. Explain the following process states:
    - New
    - Ready
    - Running
    - Waiting
    - Terminated
5. What is Process Control Block (PCB)?
6. What is process scheduling?
7. Why is CPU scheduling required?
8. Explain FCFS scheduling.
9. Explain SJF scheduling.
10. Explain Round Robin scheduling.
11. Explain Priority Scheduling.
12. Compare FCFS and Round Robin.
13. Explain context switching.
14. Explain the importance of process management.